YISHUN INNOVA JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATION
Higher 2

CANDIDATE NAME			
CG		INDEX NO	

PHYSICS 9478/03

Paper 3 Longer Structured Questions **27 August 2026**

2 hours

Candidates answer on the Question Paper.
No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

Write your name, class and index number in the spaces at the top of this page.
Write in dark blue or black pen on both sides of the paper.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, highlighters, glue or correction fluid/tape.

The use of an approved scientific calculator is expected, where appropriate.

Section A
Answer **all** questions.
Section B
Answer **one** question only.

The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiner's Use	
Paper 3	
Section A	
1	**/ 6**
2	**/ 5**
3	**/ 10**
4	**/ 10**
5	**/ 10**
6	**/ 6**
7	**/ 8**
Section B	
8	**/ 20**
9	**/ 20**
Penalty	
Paper 3 Total	
	/75

Data

speed of light in free space	$c = 3.00 \times 10^{8}\,\text{m s}^{-1}$
permeability of free space	$\mu_0 = 4\pi \times 10^{-7}\,\text{H m}^{-1}$
permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\,\text{F m}^{-1}$
	$(\frac{1}{4\pi\varepsilon_0} = 8.99 \times 10^{9}\,\text{m F}^{-1})$
elementary charge	$e = 1.60 \times 10^{-19}\,\text{C}$
Planck constant	$h = 6.63 \times 10^{-34}\,\text{J s}$
unified atomic mass constant	$u = 1.66 \times 10^{-27}\,\text{kg}$
rest mass of electron	$m_e = 9.11 \times 10^{-31}\,\text{kg}$
rest mass of proton	$m_p = 1.67 \times 10^{-27}\,\text{kg}$
molar gas constant	$R = 8.31\,\text{J K}^{-1}\,\text{mol}^{-1}$
Avogadro constant	$N_A = 6.02 \times 10^{23}\,\text{mol}^{-1}$
Boltzmann constant	$k = 1.38 \times 10^{-23}\,\text{J K}^{-1}$
gravitational constant	$G = 6.67 \times 10^{-11}\,\text{N m}^2\,\text{kg}^{-2}$
acceleration of free fall	$g = 9.81\,\text{m s}^{-2}$

Formulae

uniformly accelerated motion	$s = ut + \frac{1}{2}at^2$
	$v^2 = u^2 + 2as$
work done on / by a gas	$W = p\Delta V$
pressure	$p = \frac{F}{A}$
gravitational potential	$\phi = -\frac{GM}{r}$
temperature	$T/\text{K} = T/^\circ\text{C} + 273.15$
pressure of an ideal gas	$p = \frac{1}{3}\frac{Nm}{V}\langle c^2 \rangle$
mean translational kinetic energy of an ideal gas particle	$E = \frac{3}{2}kT$

displacement of particle in s.h.m.	$x = x_0 \sin \omega t$
velocity of particle in s.h.m.	$v = v_0 \cos \omega t = \pm \omega \sqrt{(x_0^2 - x^2)}$
electric current	$I = nAvq$
resistors in series	$R = R_1 + R_2 + ...$
resistors in parallel	$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + ...$
capacitors in series	$\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2} + ...$
capacitors in parallel	$C = C_1 + C_2 + ...$
energy in a capacitor	$U = \frac{1}{2}QV = \frac{1}{2}\frac{Q^2}{C} = \frac{1}{2}CV^2$
charging a capacitor	$Q = Q_0\left[1 - e^{-\frac{t}{\tau}}\right]$
discharging a capacitor	$Q = Q_0 e^{-\frac{t}{\tau}}$
RC time constant	$\tau = RC$
electric potential	$V = \frac{Q}{4\pi\varepsilon_0 r}$
alternating current / voltage	$x = x_0 \sin \omega t$
magnetic flux density due to a long straight wire	$B = \frac{\mu_0 I}{2\pi d}$
magnetic flux density due to a flat circular coil	$B = \frac{\mu_0 NI}{2r}$
magnetic flux density due to a long solenoid	$B = \mu_0 nI$
energy states for quantum particle in a box	$E_n = \frac{h^2}{8mL^2}n^2$
radioactive decay	$x = x_0 e^{-\lambda t}$
radioactive decay constant	$\lambda = \frac{\ln 2}{t_{\frac{1}{2}}}$

Answer **all** the questions in the spaces provided.

1 A car of mass 1800 kg is moving in a straight line. Fig. 1.1 shows the variation with time t of the momentum p of the car.

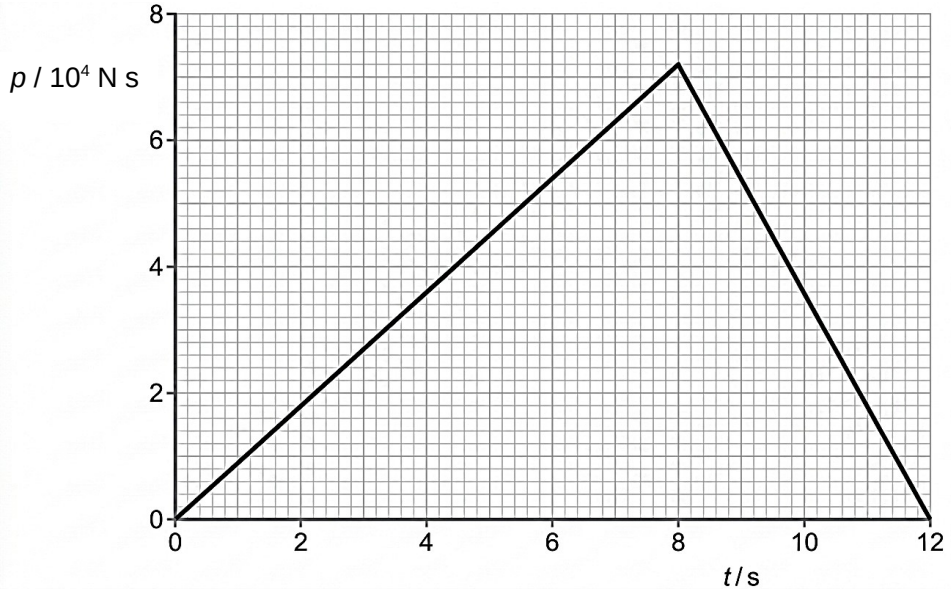


Fig. 1.1

(a) Calculate the maximum kinetic energy of the car.

kinetic energy = …….. J [2]

(b) Determine the distance travelled by the car between times t = 0 and t = 12 s.

distance = …….. m [2]

(c) On Fig. 1.2, sketch the variation with time t of the acceleration a of the car from t = 0 and t = 12 s.

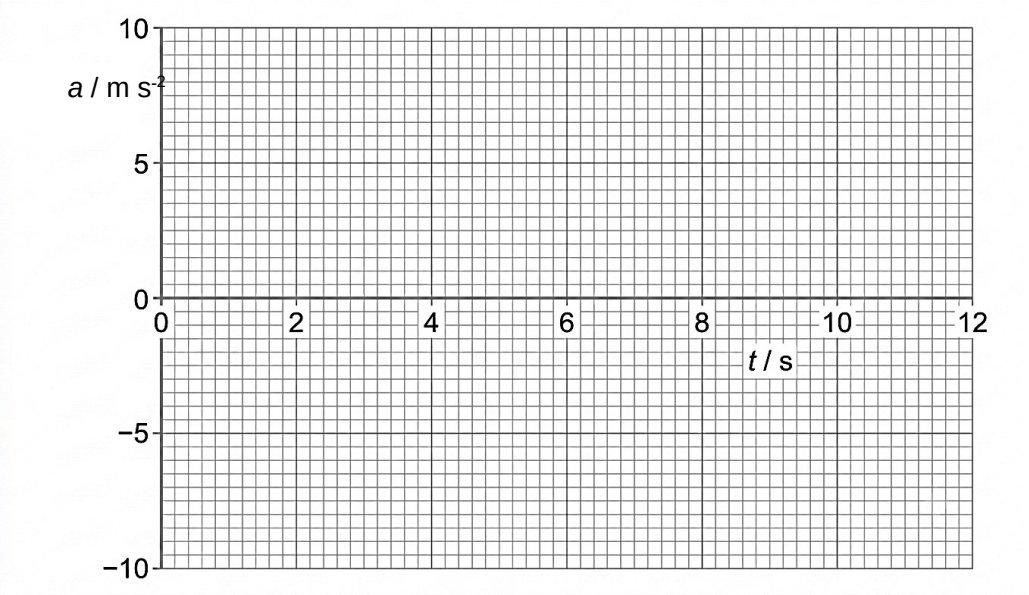


Fig. 1.2

[2]

[Total: 6]

2 A rescue drone is flying horizontally at a constant height of 80 m above a horizontal valley floor. The drone's speed u_D is 25 m s^{-1}.

The drone releases a supply package. The package falls under gravity.

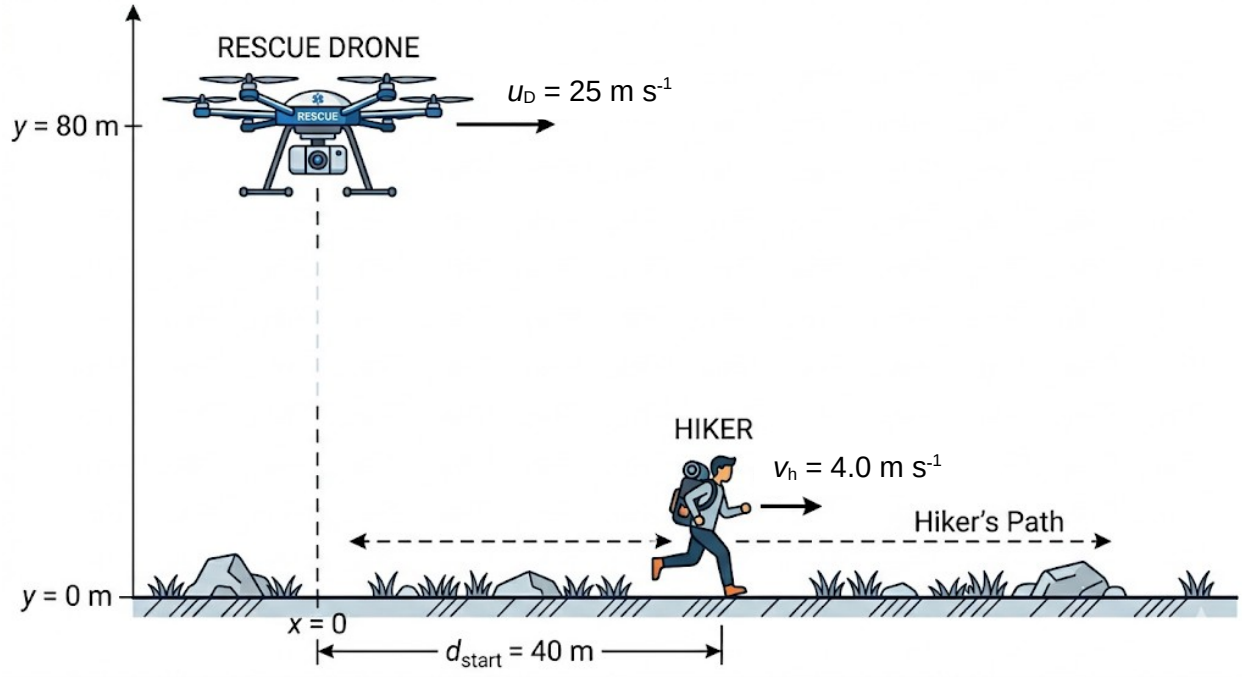


Fig. 2.1

At the instant the package is released, a hiker is horizontally in front of the drone by a distance d_{start} of 40 m (i.e., ahead along the direction of flight). The hiker immediately starts running horizontally at a velocity v_h of 4.0 m s^{-1} in the same direction as the drone's flight, attempting to catch the package.

Neglecting the effects of air resistance,

(a) calculate the time taken for the package to hit the ground,

time = …….. s [2]

(b) calculate the horizontal distance the package travels from the release point to the ground,

distance = …….. m [1]

(c) determine whether the hiker successfully catches the package. Show your working.

……

……

……

…… [2]

[Total: 5]

3 **(a)** **(i)** The gravitational potential on the surface of a planet P of radius R is $-\phi_s$. The planet can be considered to be an isolated sphere.

On the Fig. 3.1, sketch the variation of the gravitational potential with distance r from the centre of the planet for values of r between R and $4R$.

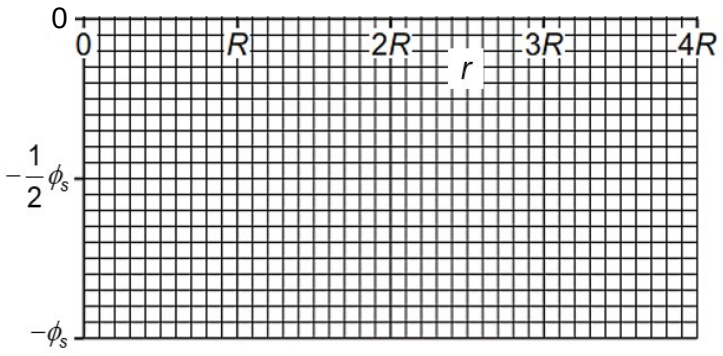


Fig. 3.1

[1]

(ii) Explain why is the gravitational potential negative.

……

……

…… [2]

(b) The variation with $\frac{1}{r}$ of the gravitational potential ϕ at different distance r from the centre of planet P is shown in Fig. 3.2.

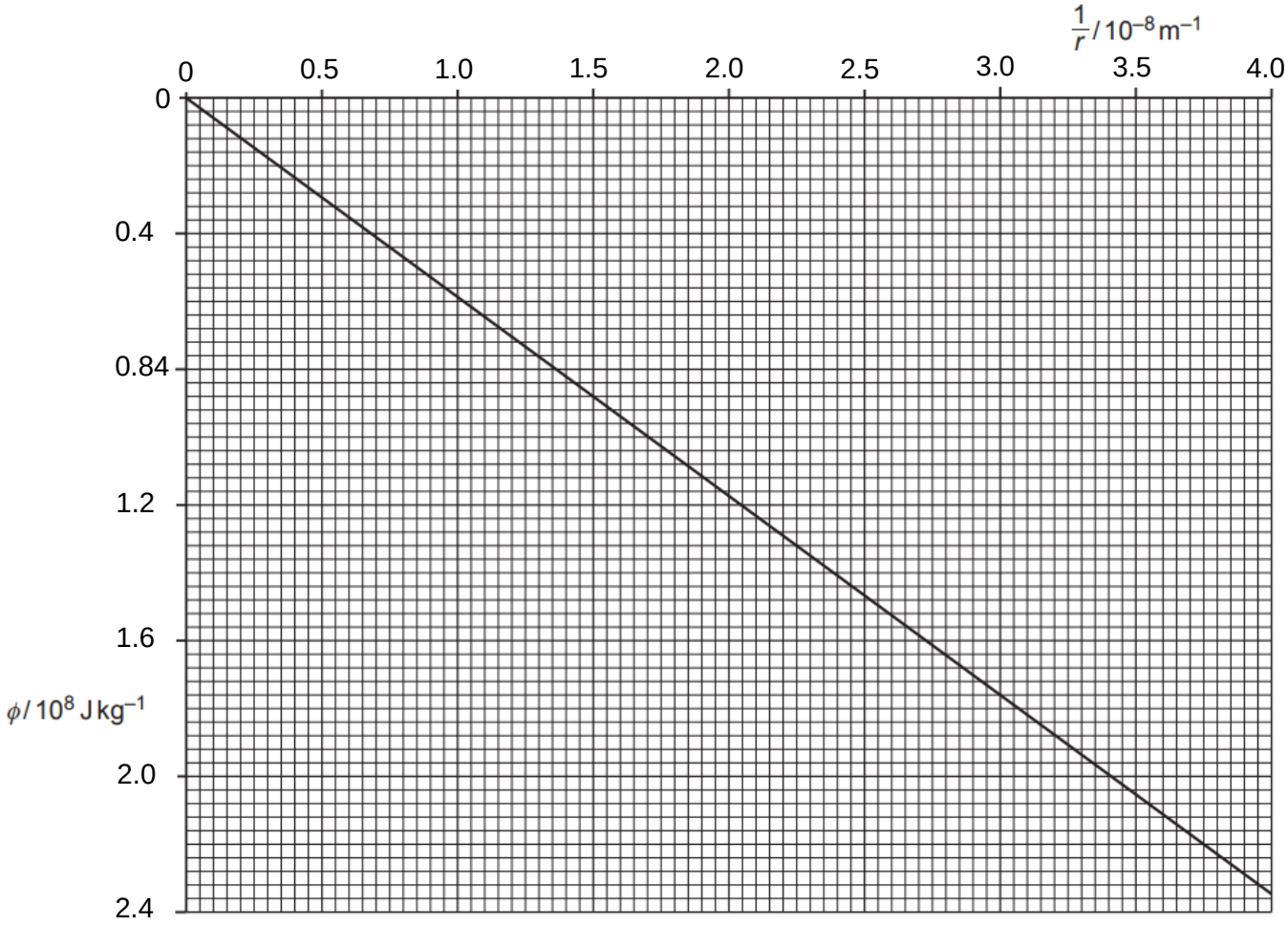


Fig. 3.2

Using Fig. 3.2, show that the mass of the planet is 8.8×10^{25} kg.

[1]

(c) **(i)** The period of rotation of the planet is 0.72 Earth days.

A satellite in orbit around the planet remains above the same point on the surface of the planet.

Determine the radius of the orbit of the satellite. Explain your working clearly.

radius = …….. m [3]

(ii) The speed of the satellite in **(c)(i)** is 8400 m s^{-1}. The mass of the satellite is 1200 kg. Determine the minimum amount of additional energy required to move the satellite from its orbit to infinity.

energy = …….. J [3]

[Total: 10]

4 **(a)** A cylinder that contains a fixed amount of ideal gas is shown in **Fig. 4.1.**

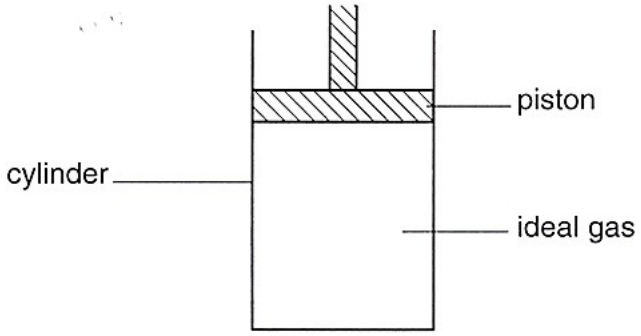


Fig. 4.1

The cylinder has a piston that can move freely.

(i) When the piston is pushed downwards slowly, keeping the temperature constant, explain how the molecules of the gas produce an increase in gas pressure.

……

……

……

…… [2]

(ii) At one instant, the gas is at a volume of 5.0×10^{-3} m^3, temperature 35 ˚C and pressure 1.0×10^5 Pa. It is then heated to a temperature of 80 ˚C at constant pressure.

Determine

1. the volume of the gas at 80 ˚C,

volume = ……..m^3 [2]

2. the work done by the gas.

work done = …….. J [1]

(b) An electric kettle, rated 230 V, 8.0 A, contains some water. It is placed on a weighing balance, as shown in Fig. 4.2.



Fig. 4.2

The kettle is switched on and, when the water is boiling, the reading on the balance decreases by 8.1 g in 10.0 s.

(i) Calculate the power rating of the kettle.

power rating = …….. W [1]

(ii) Calculate the specific latent heat of vaporisation of water.

specific latent heat = …….. J kg^{-1} [2]

(iii) Heat loss to the surroundings is one possible source of error in the determination of the specific latent heat in part **(b)(ii)**. State one other source of error and explain how it affects the calculated value of the specific latent heat.

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……

……

…… [2]

[Total: 10]

5 (a) Define electric field strength.

……

……

……[1]

(b) Fig. 5.1 shows two plates A and B, a distance 30 mm apart in vacuum, with A at a potential of -4.5 V and B at zero potential.

Electrons are emitted from B with upwards velocity, *u*.

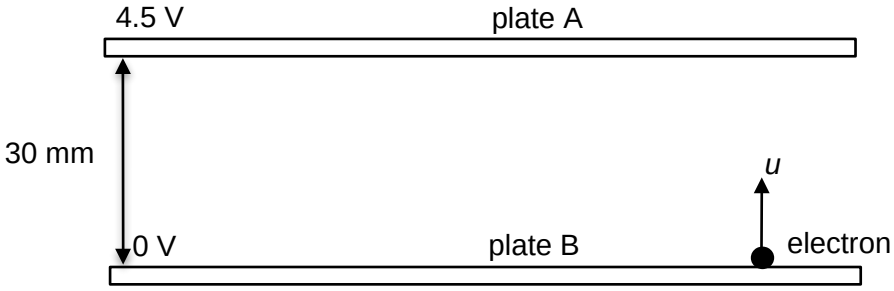


Fig. 5.1

(i) Calculate the electric field strength between the plates.

electric field strength = …….. V m^{-1} [2]

(ii) Determine the minimum value of *u* in order for electrons to reach plate A.

minimum value of *u* = …….. m s^{-1} [2]

(iii) State and explain how, if any, the value in **(b)(ii)** will change if the distance between the plates is doubled while keeping the plate potentials unchanged.

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……

……[2]

(c) An uncharged capacitor of capacitance 160 μF is charged to a potential difference of 200 V.

(i) Determine the charge on the capacitor.

charge = …….. C [1]

(ii) This charged capacitor is now connected across a resistor until the potential difference decreases to 100 V. Calculate the energy dissipated in the resistor.

energy dissipated = …….. J [2]

[Total: 10]

6 **(a)** An electron is travelling at a speed v in a straight line in a vacuum. It enters a uniform magnetic field of flux density 8.0×10^{-4} T which is acting into the plane of the paper. Initially, the electron is travelling at right angles to the magnetic field, as shown in Fig. 6.1.

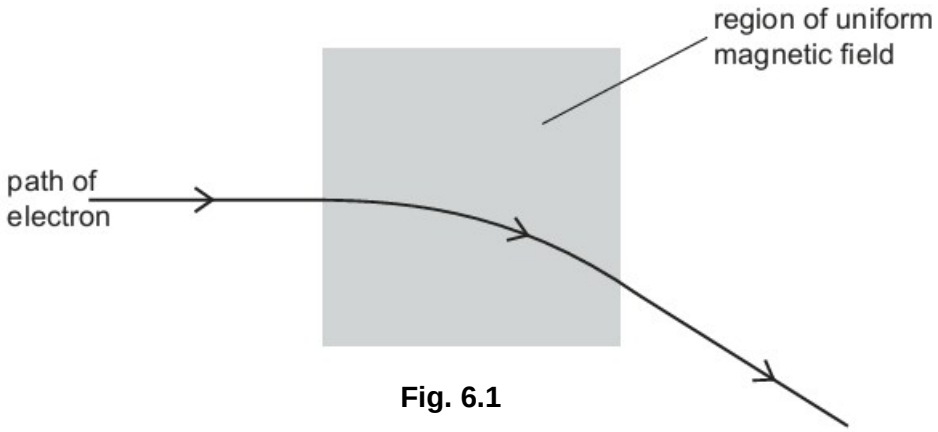


Fig. 6.1

The path of the electron in the magnetic field is an arc of a circle of radius 6.4 cm.

Determine the speed v of the electron.

speed = …….. m s^{-1} [2]

(b) A uniform electric field is now applied in the same region as the magnetic field.

The electron passes undeviated through the region of the two fields, as shown in Fig. 6.2.

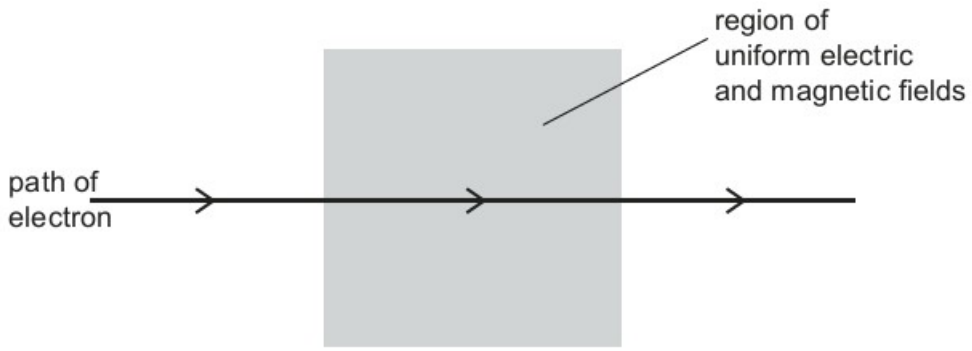


Fig. 6.2

(i) On Fig. 6.2, mark with an arrow the direction of the uniform electric field. [1]

(ii) Use the data from **(a)** to calculate the magnitude of the electric field strength.

electric field strength = …….. N C^{-1} [1]

(c) The electron in **(b)** is now replaced by a positron (which has the same mass but an equal and opposite positive charge compared to an electron) travelling at the same speed *v* along the initial path as the electron.

Describe and explain the shape of the path in the region of the magnetic and electric fields.

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…….. [2]

[Total: 6]

7 A bar magnet is suspended from the free end of a helical spring, as illustrated in Fig. 7.1.

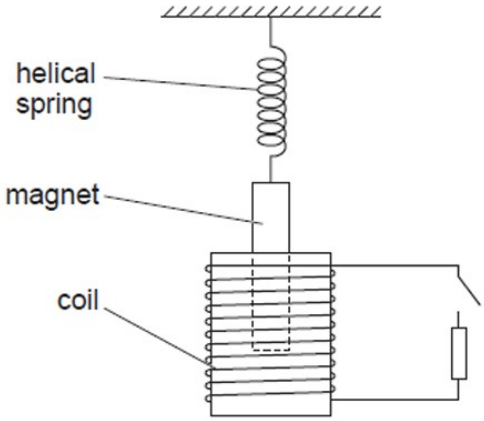


Fig. 7.1

One pole of the magnet is situated in a coil of wire. The coil is connected in series with a switch and a resistor. The switch is open.

The magnet is displaced vertically and then released. As the magnet passes through its rest position, a timer is started. The amplitude of oscillation is constant initially until time t = 3.0 s when the switch is closed.

(a) **(i)** State Faraday's law of electromagnetic induction.

……...

……..……..[2]

(ii) State Lenz's law of electromagnetic induction.

……...

……...……..[2]

(b) Use the laws of electromagnetic induction to explain why, after t = 3.0 s, the amplitude of vibration of the magnet is seen to decrease.

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……[4]

[Total: 8]

Section B

Answer **one** question from this section in the spaces provided.

8 **(a)** A loudspeaker, microphone and datalogger are arranged as shown in Fig. 8.1.



Fig. 8.1

The loudspeaker is emitting a sound wave which is detected by the microphone connected to the datalogger. The variation with time *t* of the voltage *V* of the signal captured by the datalogger as shown in Fig. 8.2.

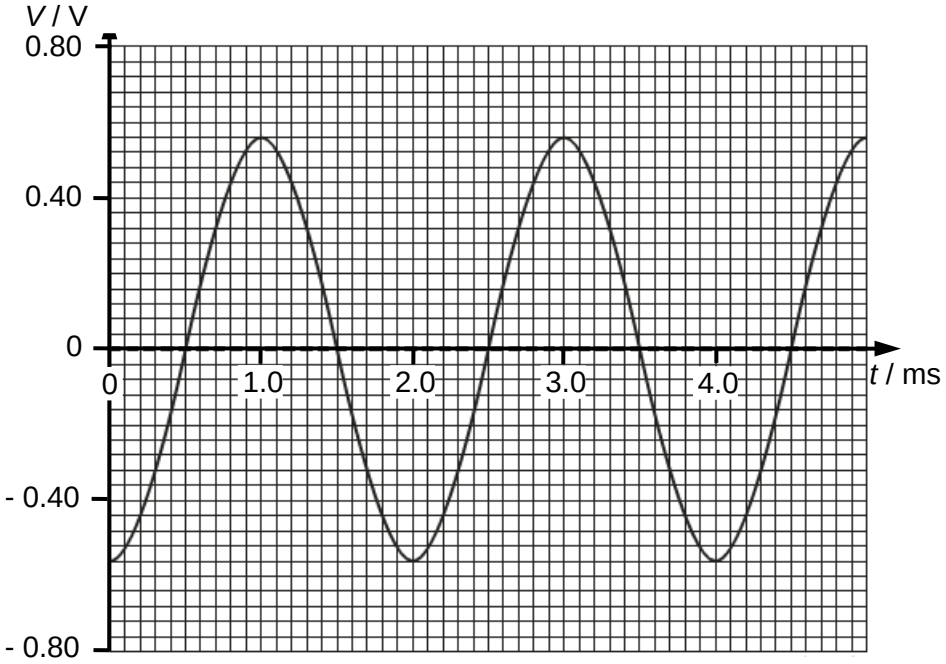


Fig. 8.2

(i) Calculate the frequency of the sound wave.

frequency = …….. Hz [1]

(ii) The intensity of the sound wave is reduced to a quarter of its original intensity without a change in frequency. Assume that the amplitude of the signal received by the datalogger is proportional to the amplitude of the sound wave.

On Fig. 8.2, sketch the trace that is now seen on the screen of the datalogger. [2]

(b) A metal sheet is now placed in front of the loudspeaker in **(a)**, as shown in Fig. 8.3.

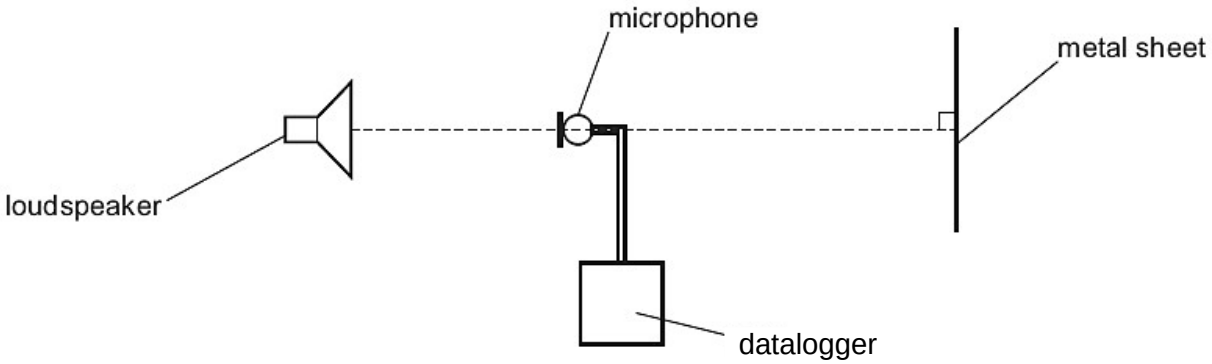


Fig. 8.3

A stationary wave is formed between the loudspeaker and the metal sheet.

The initial position of the microphone is such that the trace on the datalogger has an amplitude minimum. It is now moved a distance of 1.05 m from its initial position along the line joining the loudspeaker and metal sheet.

As the microphone moves, it passes through three positions where the trace has an amplitude maximum before ending at a position where the trace has an amplitude minimum.

(i) Explain how the stationary wave is formed from the incident sound wave.

……..

……..

……..

……..
[2]

(ii) Determine the speed of the sound in the air.

speed = …….. m s^{-1} [3]

(iii) The metal sheet in Fig. 8.3 is replaced with a material that absorbs some of the sound wave. State and explain what happens to the amplitude registered at the minimum position on the datalogger.

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……..…..
[2]

(c) An arrangement for demonstrating the interference of light is shown in Fig. 8.4.

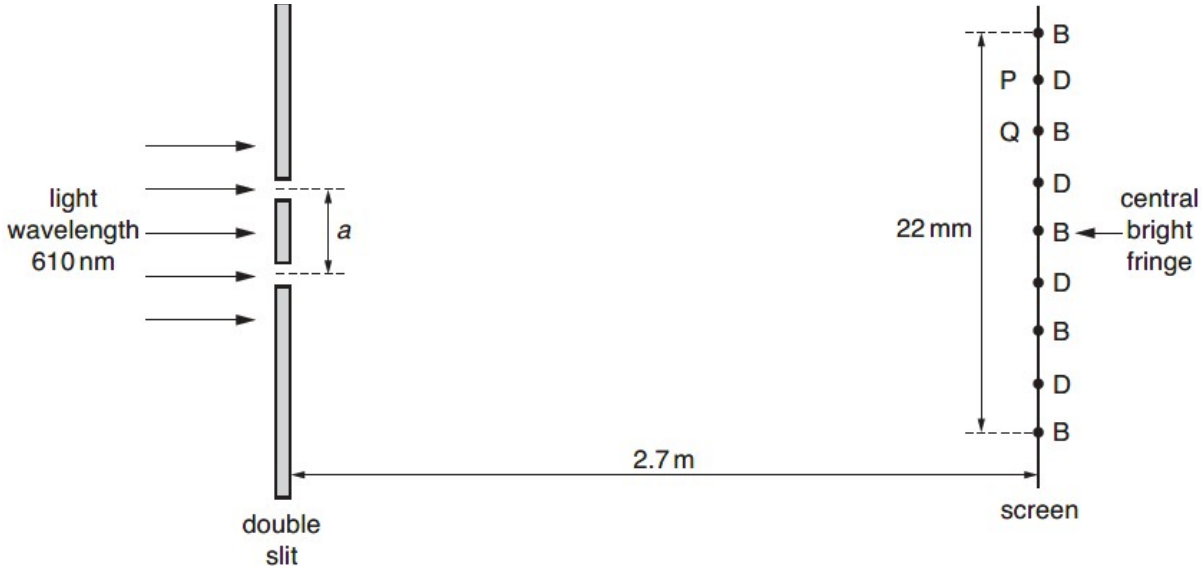


Fig. 8.4

The wavelength of the light is 610 nm. The distance between the double slit and the screen is 2.7 m.

An interference pattern of bright fringes and dark fringes is observed on the screen. The centres of the bright fringes are labelled B and centres of the dark fringes are labelled D. Point P is the centre of a particular dark fringe and point Q is the centre of a particular bright fringe, as shown in Fig. 8.4. The distance across five bright fringes is 22 mm.

(i) **1.** State the phase difference between the waves meeting at Q.

phase difference = …….. ° [1]

2. Calculate the path difference, in nm, of the waves meeting at P.

path difference = …….. nm [1]

(ii) Determine the distance *a* between the two slits.

a = …….. m [2]

(iii) The monochromatic light source of wavelength 610 nm is now replaced with a white light source.

Describe the appearance of the central fringe and the fringes observed on either side of the central fringe.

……..

……..

……..

……..… [2]

(d) A diffraction grating is used with different wavelength of visible light. The angle θ of the fourth-order maximum from the zero-order central maximum is measured for each wavelength. The variation with wavelength λ of sin θ is shown in Fig. 8.5.

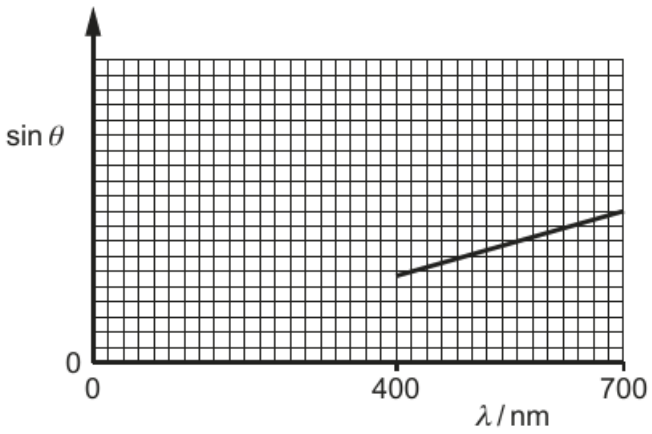


Fig. 8.5

(i) The gradient of the graph is G.

Determine an expression, terms of G, for the distance d between the centres of two adjacent slits in the diffraction grating.

d = …….. [2]

(ii) On Fig. 8.5, sketch a graph to show the results that would be obtained for second-order maxima. [2]

[Total: 20]

9 **(a)** A uranium-235 nucleus has a radius of 8.9×10^{-15} m.

Calculate, for a uranium-235 nucleus,

(i) its mass,

mass = …….. kg [1]

(ii) its mean density.

density = …….. kg m^{-3} [2]

(iii) The density of a lump of uranium is 1.9×10^{4} kg m^{-3}.

Using your answer to **(a)(ii)**, state and explain what can be inferred about the structure of the atom.

……

……

……[2]

(b) In a nuclear reaction, a possible fission reaction is

$$^{235}_{92}\text{U} + {}^{1}_{0}\text{n} \rightarrow {}^{144}_{56}\text{Ba} + {}^{90}_{36}\text{Kr} + 2\,{}^{1}_{0}\text{n}$$

.

(i) Explain what is meant by a fission reaction.

……

……

……[2]

(ii) The binding energy per nucleon of each nucleus is as follows:

$^{235}_{92}U$: 1.2191×10^{-12} J

$^{144}_{56}Ba$: 1.3341×10^{-12} J

$^{90}_{36}Kr$: 1.3864×10^{-12} J

Use these data to calculate

1. the energy released in this fission reaction (give your answer to three significant figures),

energy = …….. J [3]

2. the mass equivalent of this energy.

mass = …….. kg [2]

(iii) Suggest why the neutrons were not included in your calculations in **(ii) 1.**

……

……..[1]

(c) Strontium-90 is a radioactive isotope having a half-life of 28.0 years. Strontium-90 has a density of 2.54 g cm^{-3}.

A sample of Strontium-90 has an activity of 6.4×10^{9} Bq.

(i) Define decay constant.

……..

……[1]

(ii) Calculate

1. the decay constant λ of Strontium-90,

λ = …… s^{-1} [1]

2. the mass of Strontium-90 in the sample,

mass = …… g [3]

3. the volume of Strontium-90 in the sample.

volume = …… cm^3 [1]

(iii) By reference to your answer in **(c)(ii)3.**, suggest why dust that has been contaminated with Strontium-90 presents a serious health hazard.

……..

……..

……[1]

[Total: 20]